Database Transactions and ACID Exercises

Multiple Choice

1. A transaction debits one account, then encounters an error before crediting the other. Which ACID property requires that the failed transfer leave neither change in effect?

A. Consistency
B. Atomicity
C. Durability
D. Isolation

2. A transaction mistakenly subtracts $100 from both accounts and successfully commits both updates. Which property is violated by the incorrect combined balance?

A. Isolation
B. Durability
C. Atomicity
D. Consistency

3. In MySQL InnoDB, autocommit is enabled and a session issues `START TRANSACTION` before two `UPDATE` statements. When are both updates saved together?

A. When `COMMIT` succeeds
B. When the first `UPDATE` succeeds
C. When the second `UPDATE` succeeds
D. When the session disconnects

4. A transaction changes an account balance from $500 to $400 but hasn't committed. A second transaction reads $400. What happened?

A. A rollback
B. A durable commit
C. A dirty read
D. A deadlock

5. Power fails after a transfer's commit becomes durable, but an updated balance hasn't reached the data file. Which recovery action can restore that balance?

A. Undo the committed transfer.
B. Roll back the transfer on the original connection.
C. Repeat the transfer as a new transaction.
D. Redo the logged change.

6. Why can running all transactions one at a time reduce throughput and increase latency?

A. Serial execution requires a separate commit for each SQL statement.
B. Other transactions must wait, even when the current transaction is waiting for disk I/O or another CPU core is available.

C. Serial execution prevents the database from writing a transaction log.
D. Other transactions must repeat their reads after every commit.

True or False

1. A single `COMMIT` can save the changes made by two `UPDATE` statements in the same transaction.
2. A `ROLLBACK` issued after a successful `COMMIT` can undo that committed transaction.
3. An uncommitted update can reach a data file before the transaction commits.
4. Once an updated value reaches a data file, reading it cannot count as a dirty read.
5. A transaction can commit all its changes atomically and still violate a business rule because of a bug in its SQL.
6. During crash recovery, undo can remove changes made by a transaction that never committed.
7. A backup by itself can restore transactions committed after the backup was taken.